

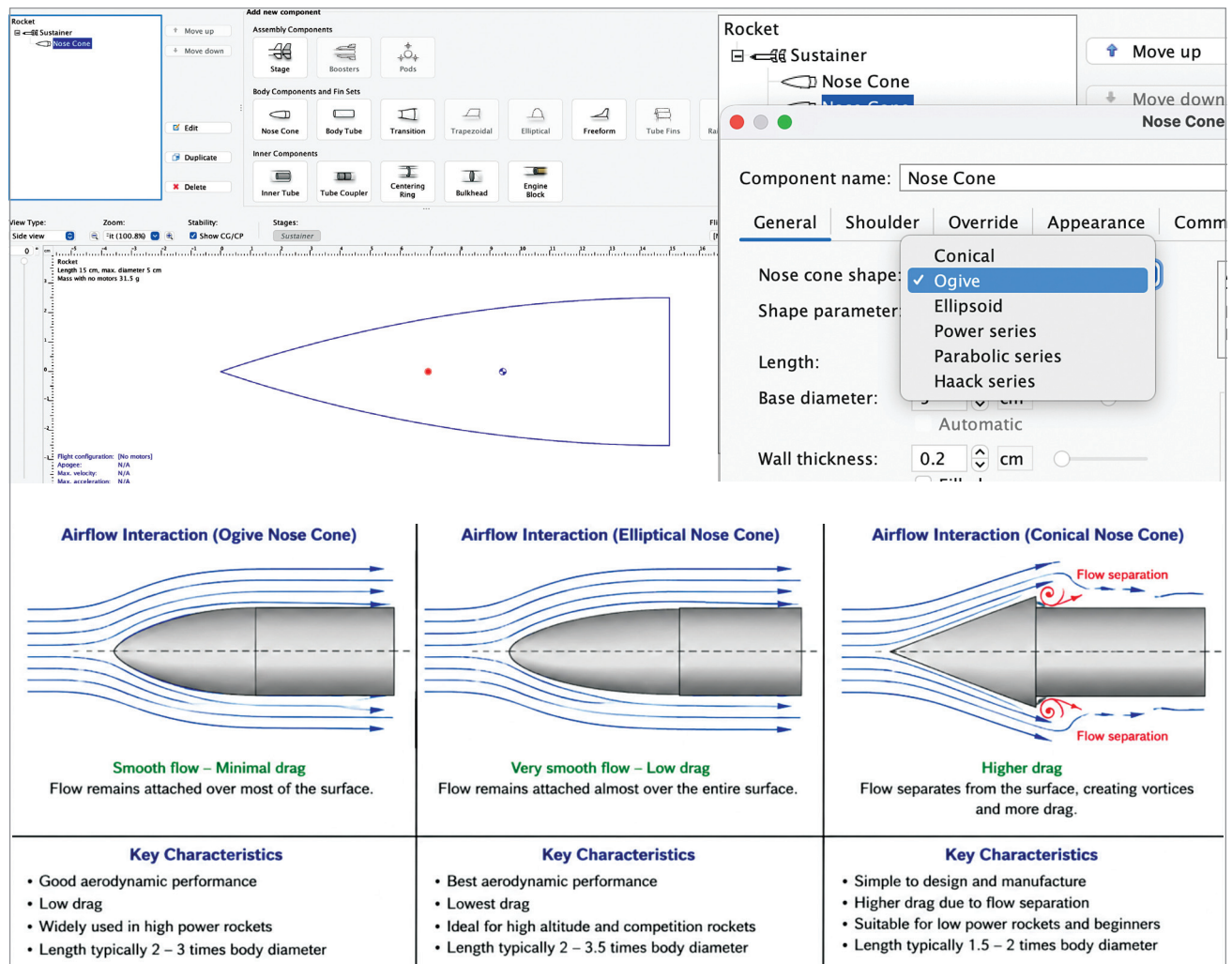


Fig. 2: Model rocket with different nose cone configurations

The upcoming parts of this series of articles will explore core rocketry concepts in greater detail, including thrust-to-weight ratio, fin shape and sizing, CG and CP, stability margin, aerodynamics, drag, and structural as well as material considerations. The series will also cover several practical aspects involved in designing a stable, efficient, and flight-ready rocket system. Fig. 1 shows the model rocket design simulator.

Note. Readers interested in discussing ideas, sharing designs, or contributing to the development process can connect using the LinkedIn QR code provided below.

The system is currently in the simulation stage, so not



all components are required at this time. However, the final components needed for the design are listed in the Bill of Materials Tables 1 and 2.

Designing

When designing a rocket, there are a few key aspects to consider. Before discussing them, however, it is first necessary to finalise the tools used to design and simulate the rocket. There are many paid and open source tools and platforms available for rocket design. This system uses open source platforms for both design and simulation, such as OpenRocket and RASAero. Both platforms are open source and supported by strong communities. This approach aligns with the objective of keeping the rocket system design freely accessible,

allowing designers to learn, build, and contribute to the rocketry ecosystem in India. The tools can be installed using the links below:

- **OpenRocket.** <https://openrocket.info>
- **RASAero.** <https://www.rasaero.com>

After downloading and setting up the platforms, the next step is the design phase. The process is carried out step by step, understanding the key design rules and applying them while designing each rocket component.

EFY note. In rocket design, for stable flight, the CG should always be positioned ahead of the CP. $CG > CP$.

The distance between the CG and CP is known as the stability